

PREDICTIVE MODELING FRAMEWORK

# RETAIL SALES FORECASTING

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# EXECUTIVE STRATEGY

An exhaustive evaluation of predictive frameworks for automated replenishment in the Guayas Region for the 2016-2017 target period.

# CORE STRATEGIC OBJECTIVES



## SCALABILITY

Handle and consolidate large-scale transaction data from an initial 5GB dataset into streamlined daily sequences.



## STRATEGIC FOCUS

Detailed analysis of demand patterns for major regional categories: Grocery I, Beverages, and Cleaning Products.



## BENCHMARKING

A comparative stress test of XGBoost, Facebook Prophet, and SARIMAX to determine the most effective replenishment engine.

# DATA STRATEGY & ENGINEERING

- ✓ **Data Cleansing:** Imputation of missing oil prices and removal of negative sales outliers.
- ✓ **Aggregation:** Translation of raw transactions into a continuous daily total sales time series.
- ✓ **Feature Engineering:** Integration of local holidays, weekly seasonality, and 2016 earthquake shocks.
- ✓ **Correlation Audit:** Excluded oil prices after analysis showed negligible predictive correlation.



# DEMAND DRIVERS & CONTEXT

## TEMPORAL PATTERNS

Weekly heartbeat detection: Friday through Sunday represents the core sales volume block.

## SYSTEMIC SHOCKS

Distinguishing one-off events (2016 Earthquake) from recurring seasonal trends for model robustness.

*"Refining feature sets for generalizable performance over 1-year horizons."*

# ENGINE 1: XGBOOST

## SUPERVISED REGRESSION

Reframed time-series as supervised learning using lags and rolling window statistics.

**Optimal Complexity: Round 738**

Best at capturing non-linear seasonal patterns and demand spikes in the Grocery I sector.





# ENGINE 2: FACEBOOK PROPHET

## ADDITIVE COMPONENTS

Specifically modeled for trend detection, holiday effects, and yearly seasonality.

- Strong overall trend tracking
- Holiday-aware prediction
- ⚠ Slight negative bias in peak detection



# ENGINE 3: SARIMAX



## STATISTICAL BASELINE

Extension of ARIMA to incorporate seasonality ( $s=7$ ) and exogenous holiday regressors.

Provided the most consistent statistical benchmark, though lacked the adaptability of gradient boosting engines.



# PERFORMANCE MATRIX

| Predictive Engine | MAE   | RMSE  |
|-------------------|-------|-------|
| Facebook Prophet  | 285.8 | 485.1 |
| XGBoost (ML)      | 288.6 | 499.3 |
| ARIMA             | 316.8 | 497.7 |

Facebook  
Prophet

Facebook Prophet

285.8

**BEST**

XGBoost

XGBoost

288.6

Second

ARIMA

ARIMA

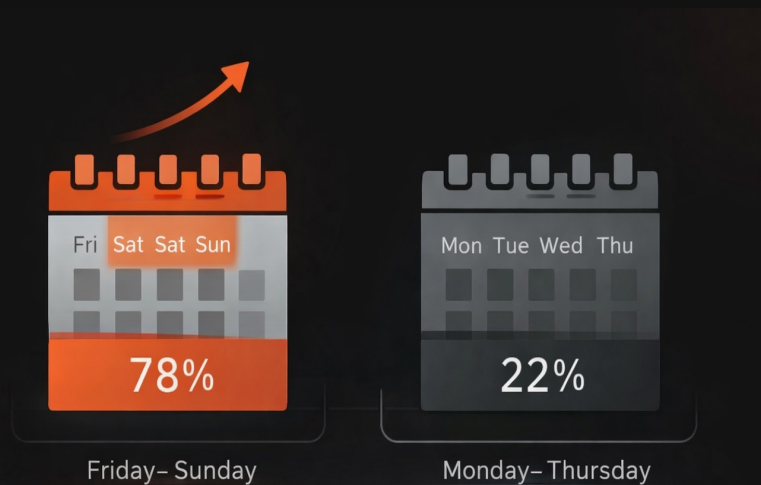
316.8

**Least accurate**

Lower MAE  
indicates higher  
precision.

Facebook Prophet  
dominates daily  
replenishment  
scenarios.

# THE WEEKLY HEARTBEAT



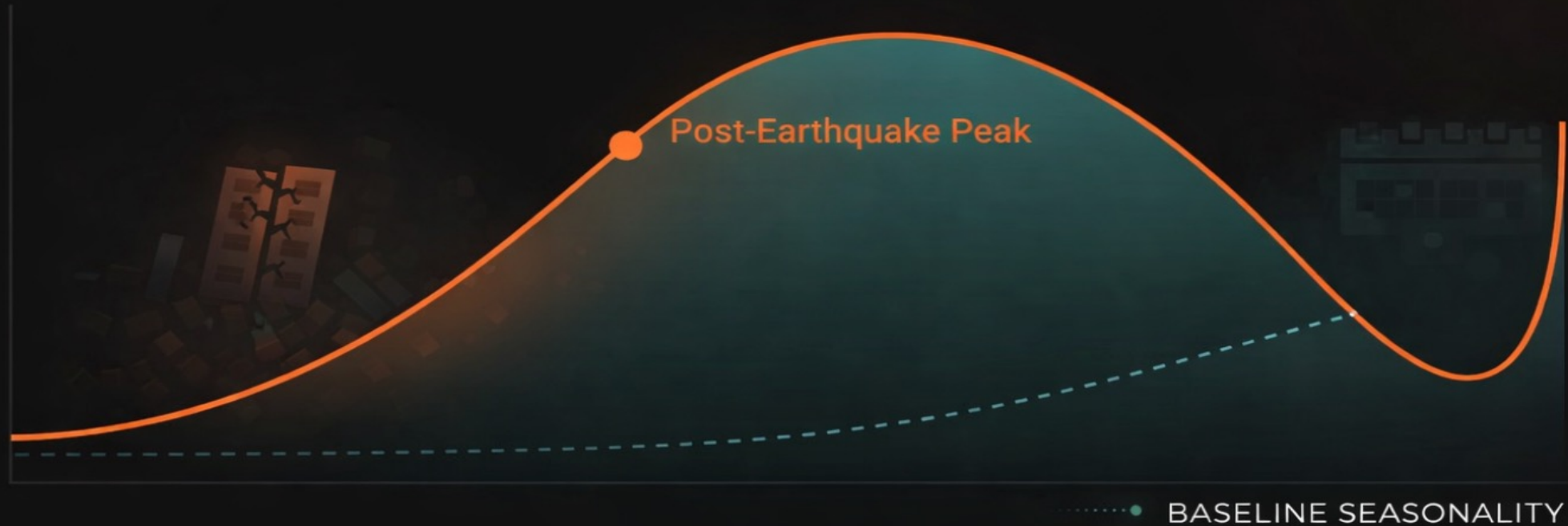
## WEEKEND CONCENTRATION

Friday-Sunday (78% Vol)

Monday-Thursday (22% Vol)

Strategic Outcome: Align regional staffing and logistics throughput with the Friday morning surge.

# EXTERNAL SHOCK RECOVERY TRENDS



**2016 Earthquake Response:** Models successfully distinguished one-off demand surges during recovery from baseline seasonality.

# IMPLEMENTATION ROADMAP



## Deploy

Prophet integration for daily regional replenishment.



## Align

Staffing schedules synchronized with weekend peaks.



## Retrain

Bi-annual modeling updates to capture shifting behavior.



## Phase 2

Integration of promotion and hyper-local marketing data.

# STRATEGIC OUTCOMES

285.8

OPTIMIZED MAE



Deploy **Prophet** for top-tier replenishment precision.



Optimize regional labor for **Friday–Sunday** spikes.



Standardize **Bi-Annual** model refresh cycle.



# NEXT PHASE: PROMOTIONAL DATA

How can external marketing catalysts further sharpen  
prediction accuracy?

Thank you for your  
attention.

I am happy to take your  
questions!

# IMAGE SOURCES



[https://png.pngtree.com/thumb\\_back/fw800/background/20260110/pngtree-modern-dark-themed-data-analytics-dashboard-with-vibrant-accent-image\\_21052087.webp](https://png.pngtree.com/thumb_back/fw800/background/20260110/pngtree-modern-dark-themed-data-analytics-dashboard-with-vibrant-accent-image_21052087.webp)

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